

OBSCYRATECH

CYBERSECURITY • TRAINING • SERVICES

PROFESSIONAL SYLLABUS

Certification Program

A foundational program designed to build an ethical hacker's mindset — covering reconnaissance, system & web hacking fundamentals, wireless and network security, defensive thinking, and career preparation for entry-level cybersecurity roles.

MODULES

12 Modules

LEVEL

**Beginner
Friendly**

FORMAT

Theory + Labs

OUTCOME

VAPT-Ready



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About the Course

This course is designed for students, aspiring cybersecurity professionals, IT support engineers, and technology enthusiasts who want to learn the fundamentals of ethical hacking in a **legal and responsible manner**. The syllabus focuses on developing an ethical hacker's mindset, understanding common attack techniques, identifying security weaknesses, and learning defensive security practices before progressing to advanced **Vulnerability Assessment and Penetration Testing (VAPT)**.

Target Audience

- Beginners interested in cybersecurity
- BCA, BSc, BE / BTech students
- IT professionals transitioning into cybersecurity
- Aspiring ethical hackers and security analysts

Prerequisites

- Basic computer knowledge
- Basic understanding of networking concepts
- Familiarity with Windows and Linux operating systems
- No prior hacking experience required

Syllabus Index

A structured pathway from cybersecurity fundamentals to a defender's mindset and a launchpad into advanced VAPT and certifications.

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Course Modules

MODULE 01 Introduction to Ethical Hacking & Cybersecurity

LEARNING OBJECTIVES

- Understand what ethical hacking is and why it is important.
- Learn legal, ethical, and professional responsibilities.
- Differentiate between hackers, attackers, and security professionals.

KEY TOPICS

- Cybersecurity fundamentals
- Types of hackers (White Hat, Black Hat, Gray Hat)
- Cyber kill chain overview
- Responsible disclosure and ethics
- Cybersecurity career paths

MODULE 02 Operating Systems for Hackers

LEARNING OBJECTIVES

- Build foundational skills in operating systems.
- Understand how attackers and defenders interact with systems.
- Gain confidence using command-line environments.

KEY TOPICS

- Windows fundamentals
- Linux fundamentals
- File systems and permissions
- Command-line basics
- User accounts and access control

MODULE 03 Networking Fundamentals for Ethical Hackers**LEARNING OBJECTIVES**

- Understand how devices communicate.
- Learn common networking concepts used in security.
- Identify network components and protocols.

KEY TOPICS

- TCP/IP and OSI models
- IP addressing and subnetting basics
- Common protocols (HTTP, HTTPS, DNS, FTP, SMTP)
- Network devices and architecture
- Packet flow fundamentals

MODULE 04 Footprinting and Information Gathering**LEARNING OBJECTIVES**

- Learn how organizations expose information publicly.
- Understand passive information gathering techniques.
- Develop reconnaissance thinking without active attacks.

KEY TOPICS

- Open-source intelligence (OSINT)
- Domain and DNS information
- Public records and metadata
- Search engine intelligence
- Social media intelligence gathering

MODULE 05 Enumeration and System Discovery Basics**LEARNING OBJECTIVES**

- Understand how systems reveal information.
- Learn safe discovery techniques.
- Identify exposed services and resources.

KEY TOPICS

- Service identification
- User and group enumeration concepts
- Shared resources discovery
- Banner grabbing basics
- Network service analysis

MODULE 06 System Hacking Fundamentals**LEARNING OBJECTIVES**

- Understand common methods used to gain unauthorized access.
- Learn password security concepts.
- Explore authentication weaknesses from a defensive perspective.

KEY TOPICS

- Password attacks fundamentals
- Password cracking concepts
- Authentication mechanisms
- Privilege escalation concepts
- Account security best practices

MODULE 07 Web Hacking Fundamentals**LEARNING OBJECTIVES**

- Understand how web applications work.
- Identify common web security weaknesses.
- Learn basic vulnerability discovery techniques.

KEY TOPICS

- Web application architecture
- Input validation issues
- SQL Injection fundamentals
- Cross-Site Scripting (XSS) basics
- Session and cookie security

MODULE 08 Sniffing and Traffic Analysis**LEARNING OBJECTIVES**

- Understand how data travels across networks.
- Learn how attackers intercept communications.
- Analyze network traffic safely.

KEY TOPICS

- Packet capture fundamentals
- Network sniffing concepts
- Protocol analysis
- Man-in-the-Middle concepts
- Secure communication practices

MODULE 09 Wireless Security Fundamentals**LEARNING OBJECTIVES**

- Understand wireless network security.
- Learn common Wi-Fi attack concepts.
- Identify secure wireless configurations.

KEY TOPICS

- Wi-Fi standards and architecture
- Wireless authentication methods
- Wireless attack concepts
- Rogue access points
- Wireless security best practices

MODULE 10 Malware, Cryptography & Social Engineering Basics**LEARNING OBJECTIVES**

- Understand common cyber threats.
- Learn how encryption protects information.
- Recognize human-focused attack techniques.

KEY TOPICS

- Types of malware (Virus, Worm, Trojan, Ransomware)
- Basic malware behavior analysis
- Cryptography fundamentals
- Hashing and encryption concepts
- Social engineering and psychological manipulation

MODULE 11 Security Monitoring and Defensive Thinking**LEARNING OBJECTIVES**

- Develop a defender's mindset.
- Learn how organizations detect threats.
- Understand incident response basics.

KEY TOPICS

- Security monitoring concepts
- Log analysis fundamentals
- Security awareness programs
- Incident response lifecycle
- Cybersecurity best practices

MODULE 12 Ethical Hacking Capstone & Career Preparation**LEARNING OBJECTIVES**

- Apply learned concepts in a controlled environment.
- Understand professional expectations.
- Prepare for entry-level cybersecurity roles.

KEY TOPICS

- Ethical hacking methodology overview
- Security assessment workflow
- Documentation fundamentals
- Professional communication skills
- Certification and career roadmap

Recommended Hands-On Labs

Throughout the course, students complete guided practical exercises to reinforce every module:

Systems & Networking

- Linux and Windows administration basics
- Network traffic observation
- Log analysis exercises

Offensive Fundamentals

- Password security demonstrations
- Web vulnerability identification labs
- Basic malware behavior observation

Wireless & Awareness

- Wireless security awareness exercises
- Security awareness simulations
- Safe traffic capture & analysis labs

Capstone Activity

- Guided end-to-end ethical hacking exercise
- Mini assessment & documentation practice
- Final presentation & peer review

Recommended Certifications

After completing this program, students are well-prepared to pursue globally recognized cybersecurity certifications:

BEGINNER LEVEL

CompTIA Security+

Microsoft Security Fundamentals

Cisco Certified Support Technician (CCST) Cybersecurity

INTERMEDIATE LEVEL**Certified Ethical Hacker (CEH)****CompTIA CySA+****eJPT (eLearnSecurity Junior Penetration Tester)**

Career Opportunities

Entry-level cybersecurity roles you can pursue after completing the Ethical Hacking Certification Program:

Security Operations

- SOC Analyst (Tier 1)
- Security Operations Trainee
- Incident Response Trainee
- Security Monitoring Analyst

Offensive Security (Junior)

- Junior Ethical Hacker
- Junior Penetration Tester
- Vulnerability Assessment Trainee
- Bug Bounty Hunter (Entry Level)

IT & Support Security Roles

- Cybersecurity Support Engineer
- IT Security Administrator
- Network Security Trainee
- Help Desk Security Analyst

Awareness & Compliance

- Security Awareness Associate
- GRC (Governance, Risk & Compliance) Trainee
- Cybersecurity Content Creator
- Security Documentation Analyst

Program Promise: This program builds a strong ethical hacking foundation aligned with industry standards and prepares students for globally recognized certifications like **CompTIA Security+**, **CEH**, **CySA+**, and **eJPT**. Every module blends theory with hands-on labs, ethics, and a defender's mindset — the perfect launchpad into advanced **VAPT** and a cybersecurity career.

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Start your journey into



Join Obscyratech's Ethical Hacking Certification Program and develop the mindset, foundational skills, and hands-on confidence needed to launch a career in cybersecurity — from SOC and security analysis to advanced penetration testing.

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LOCATION

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PROGRAM

**Ethical Hacking
Certification Program**